CS-335: AI Agents for Bioinformatics
Project Proposal

Prepared for

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By

Team Members

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Executive Summary

Our team's mission is to develop a sophisticated AI tool that helps non-bioinformatician professionals in disciplines like medicine, forensics, and pharmacology process and interpret genomic data without needing the technical skills and knowledge of trained bioinformaticians. The growing availability of genomic data has created immense potential for discovery, but accessing and interpreting this information currently requires the use of complex tools and the assistance of professional bioinformaticians. We propose developing a multi-agent AI system that enables users with little to no background in bioinformatics to work with and understand genomic data to make accurate and informed decisions in a significantly shorter amount of time.

To achieve this goal, we will design a multi-agent AI system utilizing two to three open-source large language models (LLMs), each performing specialized functions. These LLMs will read, process, and analyze genomic data with minimal human intervention. For maximal accuracy, we will choose and test at least one LLM specialized in coding. It will ultimately create the necessary Python code to read and process input genomic data. A separate LLM will be required to interpret the user's prompt and generate elaborate and meaningful instructions that the coder LLM can follow to create an effective working Python program that is then fed back into the thinking LLM for the final analysis and interpretation of the data. Finally, we will need to write a Python script that connects the two (or more) LLMs for a reasonably timed and accurate workflow.

The final deliverable of this project is a fully functioning multi-agent AI system capable of interpreting non-technical prompts and inputted genomic data and converting these interpretations to useful insights in a reasonable amount of time with minimal human intervention. The first phase of the project is building the system using the selected open-source LLMs and is on track to be completed by December 2025. The second phase is testing the system using genomic data and fine-tuning the system to enhance its performance in tasks specifically related to bioinformatics. This phase is set to be completed by March 2026.

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Section A. Problem Statement

Dealing with biological data (e.g., DNA sequences) often involves using specialized libraries for obtaining, processing, and interpreting the data, and requires assistance from a trained bioinformatician. This project is related to the field of bioinformatics and artificial intelligence, with a focus on how AI can make genomic data more accessible and easier to work with, even for individuals without bioinformatics expertise. At present, analyzing and understanding biological data, such as DNA sequences, requires specialized knowledge and complex tools. It also requires some level of programming knowledge. Professionals such as clinicians, forensic scientists, and pharmacists require assistance from bioinformatic specialists to interpret genomic data, as they lack knowledge in this field.

As genomic data continues to grow in volume and complexity, it presents a challenge for professionals seeking to utilize this data and make a change in the future. Consequently, the need for accessible analysis tools is increasing rapidly.

The problem is faced by non-bioinformaticians who need a streamlined system that would ease the process of understanding genomic data insights and perform analyses without relying on specialized bioinformaticians. Currently, most bioinformatics work requires expertise in programming, statistical analysis, and data handling, which creates a barrier for professionals who would want to work with genomic data. The unmet need is an AI-based tool that is capable of handling bioinformatics tasks and providing users with meaningful and accurate data.

The project clients are researchers, analysts, and other professionals who need to understand genomic data but lack the knowledge. The stakeholders include healthcare professionals, pharmaceutical companies, and forensic organizations.

This problem was previously addressed by the International Society for Computational Biology. They built an open-source web application, called MOLGENIS Research, to aid clinical professionals and biomedical researchers in collecting, managing, analyzing, visualizing, and sharing large and complex biomedical datasets [1]. There are currently other tools in addition to this, such as Bioperl and Biopython. However, all these tools either require manual setup or programming skills to utilize them. Therefore, there is currently no AI tool or system particularly specialized for bioinformatics tasks to process, analyze, and interpret genomic data.

The proposed solution of our project is to develop a multi-agent AI tool or system that assists non-bioinformaticians in performing bioinformatics tasks. Our solution will utilize two to three different LLMs that work together to interpret user inputs, generate the necessary Python code, execute it, and provide detailed output.

Section B. Engineering Design Requirements

The project's design focuses on creating a practical, efficient AI system for genomic analysis that operates within defined technical and budgetary constraints. It will use open-source models under permissive licenses, run on a single GPU with <= 24GB VRAM, and remain within a $1,500 budget. The system integrates a thinking agent, which interprets user prompts, and a coder agent, which autonomously generates, executes, and debugs Python code. Together, they will process standard genomic data formats (FASTA, FASTQ, VCF) accurately and efficiently, culminating in a functional end-to-end prototype that demonstrates automated analysis of a sample of datasets.

B.1 Project Goals (i.e. Client Needs)

The overall goal of this project is to empower professionals in fields such as medicine, forensics, and pharmacology to effectively utilize genomic data without requiring specialized bioinformatics training. The client needs genomic analysis to be more accessible, efficient, and interpretable through the use of advanced AI tools.

- To enable non-bioinformatician professionals to independently process and interpret genomic data.
- To streamline the bioinformatics workflow by reducing reliance on specialized human experts.
- To reduce the time and effort required for genomic data analysis while maintaining accuracy and reliability.
- To improve accessibility to genomic insights through an intuitive, AI-assisted interface.
- To demonstrate the feasibility of using a multi-agent AI system to support real-world decision-making based on biological data.

B.2 Design Objectives

The goal of the project is to explore creating an AI Agent that will enable non-bioinformatics users to analyze and interpret genomic data efficiently without requiring specialized technical knowledge. By leveraging open-source large language models (LLMs), the system will function as a multi-agent AI capable of processing, coding, and interpreting genomic datasets automatically.

- The design will enable non-bioinformatician professionals to analyze and interpret genomic data without requiring specialized technical knowledge allowing them to make informed decisions efficiently
- The design will utilize open-source large language models (LLMs) to build a functional multi-agent AI system capable of processing, coding, and interpreting genomic datasets.

- The design will reduce the time required for data processing and interpretation by at least a 50% compared to traditional bioinformatic workflows, as measured through benchmarking testing.
- The design will generate and execute Python code automatically for bioinformatic tasks with at least 95% success in debugging and completion without human intervention.
- The design will accurately interpret user prompts and convert them into executable coding tasks with >= 90% accuracy, verified through standardized test inputs.
- The design will support standard genomic data formats, including FASTA, FASTQ, and VCF, ensuring broad compatibility with existing datasets.
- The design will deliver a working prototype by the end of the Spring 2025 semester, demonstrating an end-to-end automated analysis of sample genomic data.

B.3 Design Specifications and Constraints

The design will use open-source models within budget and hardware limits, integrating a thinking agent to interpret prompts and a coder agent to autonomously generate and execute code. It must handle standard genomic formats accurately and perform full analyses efficiently, producing a functional end-to-end prototype.

- Design must use an open source model that is released under open source licences, while avoiding Chinese models.
- Total expenditures for computing resources, storage, and software must not exceed $1,500 USD.
- Each model must not have no more than 20 billion parameters and be capable of running on a single GPU with <= 24 GB VRAM, meeting our server's available hardware limits.
- The coder agent must be able to generate, execute, and debug code without human assistance a majority of the time.
- The thinking agent must correctly interpret the user prompts and produce actionable instructions for the coder agent majority of the time, and be verified against predefined test cases.
- The integrated system must complete a full data analysis within a TBD amount of time.
- The system must be able to read and process standard genomic formats such as FASTA, FASTQ, and VCF without data loss or formatting errors.

Section C. Scope of Work

This project will deliver a multi-agent AI system for professionals in different industries to process and interpret genomic data without the assistance of professional bioinformaticians. The scope of work encapsulates the process of defining the deliverables along with the timeline expected to complete the project. Currently, the only resource needed for the first phase is access to VCU's GPU. This is subject to change as we dive deeper into the project.

C.1 Deliverables

To ensure the timeliness of implementing the project by the specified deadlines, we have set the project's deliverables as follows:

- A program capable of interpreting simple prompts, reading and processing genomic datasets, and analyzing the data with minimal human intervention to generate meaningful insights.
- A reliable coder LLM capable of generating accurate and functional python scripts.
- A sophisticated thinker LLM capable of interpreting non-technical prompts to generate elaborate instructions for the coder LLM and communicating the insights effectively to the user.
- A python based program that connects selected LLMs to ensure smooth and timely workflow.
- Required academic deliverable including team contract, project proposal, preliminary design report, fall poster and presentation, final design report, and Capstone EXPO post and presentation.

C.2 Milestones

Ensuring the timeliness of the project also includes defining explicit milestones with realistic deadlines. For this project, we have defined the milestones as follows:

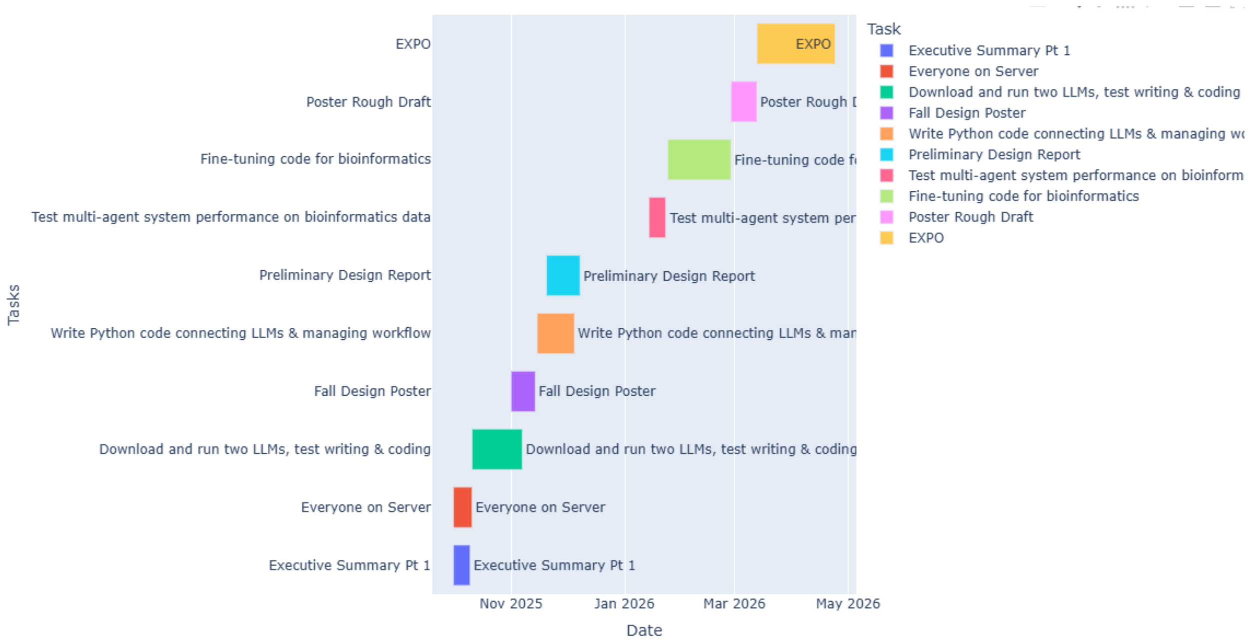
- Submit preliminary Project Proposal: 10/10/2025
- Ensure full access to the server by all team members: 10/11/2025
- Select and test coder and thinker LLMs: 11/7/2025
- Submit Fall Design Poster: 11/14/2025
- Write Python code that connects LLMs and manages workflow: 12/5/2025
- Preliminary Design Report: 12/8/2025
- Test multi-agent system performance on genomic datasets: 1/23/2026

- Refine Python script for enhanced performance in bioinformatics related tasks: 2/27/2026
- Submit Poster Rough Draft: 3/13/2026
- EXPO: 4/24/2026

C.3 Resources

The resources list is subject to change as we progress through the project and experiment with the different LLM's for the specified functionalities required to build a sophisticated AI-based system. Currently, access to the VCU GPU is the only needed resource for the completion of the project.

Appendix 1: Project Timeline



Appendix 2: Team Contract (i.e. Team Organization)

Step 1: Get to Know One Another. Gather Basic Information.

Team Member Name	***Strengths each member bring to the group***	***Other Info***	***Contact Info***
Carter Struck	*Communicative, straight forward with issues, deadline conscious*	*Experience with python java react native, web scraping.*	*struckcm@vcu.edu*
Mack Hicks	*Eager to learn, work with others in incorporating each team members visions*	*Java, Python, Microsoft Power Apps, SQL*	*Hicksm5@vcu.edu*

Sona James	*Organization, time management, and problem-solving*	*I prefer working ahead of time so the team isn't rushed later on.*	*jamess9@vcu.edu*
Ozi Jawad	*Open to ideas, curious, communicative*	*Java, Python, R, sql, tableau*	*jawado@vcu.edu*

Other Stakeholders	***Notes***	***Contact Info***
Tomasz Arodz	*Friendly, communicative, open to ideas*	*tarodz@vcu.edu*

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Culture Goals	***Actions***	***Warning Signs***
Attend meetings	- *Set up reminders through a shared calendar or texting* - *Let other people know before hand if times don't work or unable to attend*	- *Student misses the first meeting, a warning is granted* - *Student misses meetings afterwards – the issue is brought up with the faculty advisor*
Be open about delays in work progress	- *Stay up to date with each other's project responsibilities* - *Set reasonable deadlines and note when an extension is needed* - *Be as open where you stand with the work as much as possible*	- *Student shows up for weekly meeting with no considerable work done*
Be open to compromise and be respectful of others' decisions.	- *Your idea did not get picked, don't crash out.*	- *Starts slacking after the idea didn't get picked.*

Step 3: Time Commitments, Meeting Structure, and Communication

Meeting Participants	***Frequency Dates and Times / Locations***	***Meeting Goals Responsible Party***

Students Only	*On Discord as needed, most likely on the weekends*	*Actively work on project (Sona will document these meetings by taking photos of whiteboards, physical prototypes, etc, then post on Discord and update Capstone Report)*
Students Only	*Every Tuesday after class*	*Update group on day-to-day challenges and accomplishments*
Students + Faculty advisor(Same for sponsor	*Meet via zoom Wednesday or Monday every other week.*	*Update faculty advisor and get answers to our questions and share updates*

Step 4: Determine Individual Roles and Responsibilities

Team Member	***Role(s)***	***Responsibilities***
Mack Hicks	*System Engineer*	*Introducing design ideas* *Overseeing the specifications of the project being met*
Ozi Jawad	*Logistics Manager*	*Communicates with advisor* *Coordinates internal and external interactions* *Obtains information for the team* *Manage resource usage*
Carter Struck	*Project manager*	*Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.* *Decider in group differences.*
Sona James	*Documentation Manager*	*Documents meeting minutes - key points and decisions made during the meeting.* *Oversees statistical analysis* *Creates and maintains task lists/Organizes workspace*

Step 5: Agree to the above team contract

Team Member: Ozi Jawad *Signature: Ozi Jawad*

Team Member: Mack Hicks *Signature: Mack Hicks*

Team Member: Sona James *Signature: Sona James*

Team Member: Carter Struck *Signature: Carter Struck*

References

[1] van der Velde, K. J., Imhann, F., Charbon, B., Pang, C., van Enckevort, D., Slofstra, M., Barbieri, R., Alberts, R., Hendriksen, D., Kelpin, F., de Haan, M., de Boer, T., Haakma, S., Stroomberg, C., Scholtens, S., van de Geijn, G.-J., Festen, E. A., Weersma, R. K., & Swertz, M. A. (2018). Molgenis Research: Advanced Bioinformatics Data Software for non-bioinformaticians. Bioinformatics, 35(6), 1076–1078. https://doi.org/10.1093/bioinformatics/bty742